METHODOLOGY DOCUMENT
MSCI SUSTAINABILITY AND CLIMATE

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MSCI ESG Ratings Methodology: Climate Change Vulnerability Key Issue

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Introduction

This document provides essential information on the components of the Climate Change Vulnerability Key Issue to enable users of ESG Ratings to understand how MSCI Sustainability & Climate's (MSCI S&C's)[1] outputs are determined. Climate Change Vulnerability is a Key Issue in the Environmental Pillar of the MSCI ESG Ratings model. Insurance companies are assessed on the physical risk that climate change may pose to insured assets or individuals.

For additional details on the MSCI ESG Ratings Model, refer to the Data, ratings and scores section of "ESG Ratings Methodology."

Exhibit 1: MSCI ESG Key Issue hierarchy

3 Pillars	10 Themes	33 ESG Key Issues
Environmental	**Climate Change**	Carbon Emissions
		Climate Change Vulnerability
		Financing Environmental Impact
		Product Carbon Footprint
	Natural Capital	Biodiversity & Land Use
		Raw Material Sourcing
		Water Stress
	Pollution & Waste	Electronic Waste
		Packaging Material & Waste
		Toxic Emissions & Waste
	Environmental Opportunities	Opportunities in Clean Tech
		Opportunities in Green Building
		Opportunities in Renewable Energy
Social	Human Capital	Health & Safety
		Human Capital Development
		Labor Management
		Supply Chain Labor Standards
	Product Liability	Chemical Safety
		Consumer Financial Protection
		Privacy & Data Security
		Product Safety & Quality
		Responsible Investment
	Stakeholder Opposition	Community Relations
		Controversial Sourcing
	Social Opportunities	Access to Finance
		Access to Health Care
		Opportunities in Nutrition & Health
Governance	Corporate Governance	Board
		Pay

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3 Pillars	10 Themes	33 ESG Key Issues
		Ownership & Control
		Accounting
	Corporate Behavior	Business Ethics
		Tax Transparency

Risks associated with this Key Issue

- Climate model risk: inadequate climate risk modelling may underestimate or misrepresent future climate-related losses and reserving assumptions.
- Liquidity and solvency risk: timing and size of claims impact cash flows and solvency.
- Catastrophe risk: increase in frequency and severity and decrease in predictability of catastrophe losses.

Climate Change Vulnerability Key Issue score

The Climate Change Vulnerability Key Issue score evaluates the company's level of exposure to, and management of, risks on this Key Issue. The Key Issue score is based on the Exposure score and the Management score using the formula below. The Exposure score and the Management score are combined such that a higher level of exposure requires a higher level of demonstrated management capability in order to achieve the same overall Key Issue score.

$$KI_i = 7 - (\max(EXP_i, 2) - MGMT_i)$$

Where:

- KI_i is the Key Issue score for company i.
- EXP_i is the Exposure score for company i.
- $MGMT_i$is the Management score for company i.

Climate Change Vulnerability Management framework

The table below outlines the indicators mapped to their respective components that inform the Climate Change Vulnerability Management score, excluding controversies. MSCI S&C assigns each indicator a default weight. Indicator weights may be adjusted at the company level to reflect the relevance of the indicator to the company's business activities. If a company has no exposure to the business activities covered by an indicator, the indicator is assigned a weight of zero.

Component	Sustainability Indicator

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Climate Change Vulnerability Practices score	Recognition that physical risk of climate change may increase underwriting risks
	Research or exercise to quantify the impact of physical risk on insurance underwriting
	Evidence of using climate change models
Climate Change Vulnerability Performance score	Underwriting performance based on combined ratio

Climate Change Vulnerability Management score

The Climate Change Vulnerability Management score evaluates the company's ability to manage its exposure to risks on this Key Issue. It is based on weighted average of the score associated with each relevant indicator.

This score is derived from indicators that are scored on a 0-10 scale, with 10 corresponding to best practice. Refer to the Analyzing risk management section of "ESG Ratings Methodology" for additional details. Sources are company disclosures except where otherwise indicated.

The following formula is used to calculate the Climate Change Vulnerability Management score:

$$MGMT_i = \sum Indicator_Score_{j,i} \ * \ Indicator_Weight_{j,i}$$

Where:

- $MGMT_i$ is the Management score - Excluding Controversies for company i.
- $Indicator_Score_{j,i}$ is the score of indicator j for company i.
- $Indicator_Weight_{j,i}$ is the weight of indicator j for company i.

The following formula is used to calculate the scores for the components

$$COMP_{c,i} = \sum_{j \in c} Indicator_Score_{j,i} \ * \ Indicator_Weight_Scaled_{j,i}$$

Where:

- $COMP_{c,i}$ is the component score for component c, for company i.
- $Indicator_Score_{j,i}$ is the score of indicator j for company i.
- $Indicator_Weight_Scaled_{j,i}$ is the scaled weight of indicator j for company i.
- $j \in c$ indicates that the summation is over all indicators belonging to component c.

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The following formula is used to calculate the weights for the components

$$COMP_Weight_{c,i} = \sum_{j \in c} Indicator_Weight_{j,i}$$

Where:

- $COMP_Weight_{c,i}$ is the component weight for component c for company i.
- $Indicator_Weight_{j,i}$ is the weight of indicator j for company i.
- $j \in c$ indicates that the summation is over all indicators belonging to component c.

The definitions of the data points in each component are given below:

Climate Change Vulnerability Practices score

The Climate Change Vulnerability Practices score evaluates the company's initiatives to integrate climate change considerations into insurance underwriting, including use of climate models, research on physical risk impacts and recognition of increased risk exposure. This score is a component of the Climate Change Vulnerability Management score (excluding controversies) and considers attributes of the company's policies and initiatives. It is expressed as a 0–10 score, where 10 indicates strong initiatives.

- **Recognition that physical risk of climate change may increase underwriting risks**

Definition: Indicates whether the company recognizes that the physical risk of climate change may increase underwriting risks.

- **Research or exercise to quantify the impact of physical risk on insurance underwriting**

Definition: Indicates whether the company conducts research or exercises to quantify the impact of physical risk on insurance underwriting, either directly (via primary research) or indirectly (through an industry consortium).

- **Evidence of using climate change models**

Definition: Indicates whether the company uses climate change models to inform underwriting decisions.

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Climate Change Vulnerability Performance score

The Climate Change Vulnerability Performance score evaluates the company's underwriting performance, based on its three-year average combined ratio. The combined ratio measures an insurer's underwriting profitability by comparing total claims and expenses to earned premiums. It is expressed as a 0-10 score, where 10 indicates the highest level of profitability.

- **Underwriting performance based on combined ratio**

Definition: Assesses the company's underwriting performance, based on its three-year average combined ratio. The combined ratio measures an insurer's underwriting profitability by comparing total claims and expenses to earned premiums.

Climate Change Vulnerability Exposure score

The Climate Change Vulnerability Exposure score evaluates the company's exposure to risks on this Key Issue. It is based on the average of the Business and Company-Specific Exposure scores, combined with the Geographic Exposure score. The Geographic Exposure score functions as a multiplier on the averaged Business and Company-Specific Exposure scores, with an impact ranging from -50% to +50%. The Business Exposure score, Geographic Exposure score and Company-Specific Exposure score are scored on a 0-10 scale, with 10 corresponding to the highest risk and 0 corresponding to the lowest risk. Refer to Section 3.2, Analyzing risk exposure, of "ESG Ratings Methodology" for additional details. The following formula is used in the Exposure score calculation:

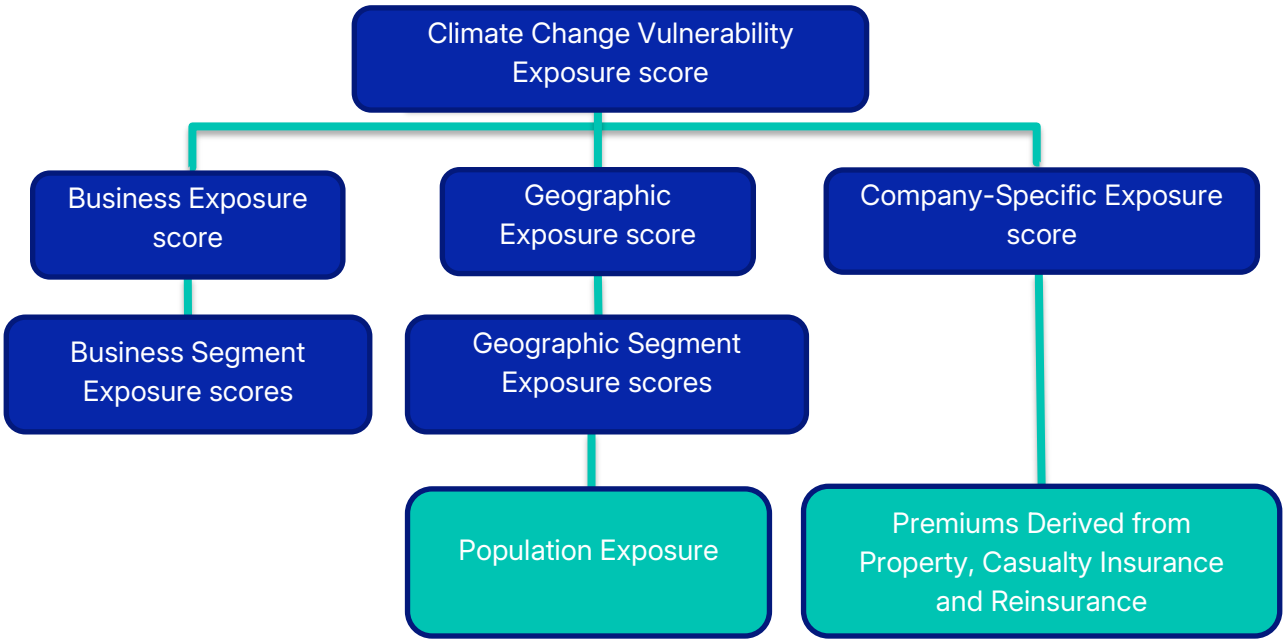
$$EXP_{CCV,i} = \left(\frac{BUS_{CCV,i} + COM_{CCV,i}}{2}\right)(1 + 0.1(GEO_{CCV,i} - 5))$$

Where:

- $EXP_{CCV,i}$ is the Climate Change Vulnerability Exposure score of company i.
- $BUS_{CCV,i}$ is the Business Exposure score of company i.
- $COM_{CCV,i}$ is the Company-Specific Exposure score of company i.
- $GEO_{CCV,i}$ is the Geographic Exposure score of company i.

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Exhibit 2: Exposure score components



Business Exposure score

The Business Exposure score is a weighted average of the Business Segment Exposure scores of a company's business segments. Scores are weighted by the proportion of a company's total revenue in each business segment, using the following formula to calculate the Business Exposure score:

$$BUS_{CCV,i} = \sum_{j=1}^{n_i} w_{Revenue,i,j} BSE_{CCV,j}$$

Where:

- $BSE_{CCV,j}$ is the Business Segment Exposure score for business segment j.
- $w_{Revenue,i,j}$ is the weight of business segment j for company i based on the contribution to total company revenue.
- n_i is the number of business segments of company i.

Business Segment Exposure scores

The Business Segment Exposure score of a business segment is based on a mapping of the business segment as disclosed by the company to a corresponding business activity. MSCI S&C uses the Standard Industrial Classification (SIC) system along with industry-specific adjustments to define business activities. The score associated with a business activity is used to calculate a Business

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Segment Exposure score. Each business activity score is determined by a qualitative assessment of whether a business activity has a core involvement in insuring property, assets and individuals for both primary insurance and reinsurance. A higher involvement will result in a higher Business Segment Exposure score for the business activity.

Geographic Exposure score

The Geographic Exposure score is the weighted average of the Geographic Segment Exposure scores of the countries, markets and regions in which a company operates. Scores are weighted by the proportion of a company's total revenue in each geographic segment using the following equation:

$$GEO_{CCV,i} = \sum_{l=1}^{n_i} w_{Revenue,i,r} GSE_{CCV,r}$$

Where:

- $GSE_{CCV,r}$ is the Geographic Segment Exposure score in region r.
- $w_{Revenue,i,r}$ is the weight of total revenue in region r for company i.
- n_i is the number of geographic segments for company i.

For geographic segments reported as regions (example: Asia Pacific), a nominal GDP-weighted country aggregation is used to calculate region-level scores, using the following equation:

$$GSE_{CCV,r} = \sum_{c=1}^{n_r} w_{GDP,r,c} GSE_{CCV,c}$$

Where:

- $GSE_{CCV,c}$ is the Geographic Segment Exposure score of country or market c.
- $w_{GDP,r,c}$ is the specific weight of country or market c within region r.

Geographic Segment Exposure scores

The Geographic Segment Exposure score of a geographic segment is determined by selecting the maximum population exposure to any natural hazard, including climate-induced natural hazards, as per the INFORM Risk Index:

$$GSE_{CCV,r} = \max\left(HPE_{k,r}\right)$$

Where:

- $GSE_{CC,r}$ is the Geographic Segment Exposure score for region r.

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- $HPE_{k,r}$ is the Hazard Population Exposure score to natural hazard k in region r.

The Hazard Population Exposure score is derived from the INFORM Risk Index, which uses estimated population exposure to each natural hazard per country or market. These hazards include earthquakes, floods, tropical cyclones, droughts, tsunamis and epidemics.

Company-Specific Exposure score

The Company-Specific Exposure score is determined by a simple average of the Property & Casualty Insurance Line score and the Reinsurance Line score:

$$COM_{CCV,i} = \frac{P\&C_i + ReINS_i}{2}$$

Where:

- $COM_{CCV,i}$ is the Company-Specific Exposure score for company i.
- $P\&C_i$ is the Property and Casualty Insurance Line score for company i.
- $ReINS_i$ is the Reinsurance Line score for company i.

The Property and Casualty Insurance Line score is calculated from the percentage of premiums derived from property and casualty insurance lines. The Reinsurance Line score is calculated from the percentage of premiums derived from reinsurance lines.

Data sources

The following sources are used to determine the Climate Change Vulnerability Exposure score:

INFORM Risk Index, Inter-Agency Standing Committee and European Commission; Refinitiv; MSCI S&C; company disclosures.

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